Joseph Walton

Email: jwgwalton@gmail.com
Telephone: 07817685781
LinkedIn: linkedin.com/in/joseph-walton-2a508591
GitHub: github.com/jwgwalton

Professional Summary

Experienced machine learning engineer with over 10 years of expertise in machine learning, AI engineering & software engineering. Demonstrable success in building and deploying advanced AI models and production-grade ML systems with significant business value. Passionate about developing secure, responsible AI systems. Possesses excellent stakeholder management skills & ability to guide organisations through AI transformation.

Technical Skills

- **Programming Languages:** Python, SQL, Java
- **Agent ecosystem:** AzureOpenAI, Huggingface, Langchain, Langgraph
- **Machine Learning Frameworks:** TensorFlow, Pytorch, Scikit-learn
- **Data Analysis:** Apache Spark, Pandas, Polars, NumPy, Matplotlib, Seaborn
- **Tools:** Git, Docker, Jupyter, MLFlow
- **Cloud Platforms:** Azure, AWS, Google Cloud Platform
- **Databases:** MySQL, PostgreSQL, Elasticsearch

Professional Experience

IBM

Lead Data Scientist
June 2025 - Present
Contract role with central government.

- Rescuing failing AI projects through implementing best practices. (Evaluations, Observability)
- Designed and delivered a data processing LLM agent saving 2 hours per case worker per day for a saving of ~£2M a year, using Azure AI Foundry, Azure Functions and LangGraph; validated with classical ML metrics by reframing it as a data aggregation and labelling tool for human verification rather than a summarisation task.
- Wrote a Claude Skill to optimise Human-in-the-Loop data ingestion for writing Evals for agent products, time saved has been ~1 hour per case in the Eval suite.
- Rewrote a data matching algorithm in Polars from Apache Spark, curating a representative dataset to validate the rewrite with classical ML metrics and optimise both model performance and processing speed. This reduced cloud costs by ~£10k per month, reducing run time from 2 hours to 10 minutes.

Intellectual Property Office

Senior Machine Learning Engineer

November 2021 - April 2025

Developing NLP solutions with a focus on improving outcomes of trade mark applications.

- Developed a product roadmap for AI applications in the IP space
- Collaboration with multiple international IP offices communicating how the UK IPO is implementing AI in the IP space, to both technical & non-technical stakeholders.
- Member of cross-government AI working group, contributing to the development of AI policy and best practices.
- Developed Generative AI powered applications within a service-oriented architecture using Langchain, AzureOpenAI & Azure Functions.
- Built a RAG agent to help novice users identify relevant Goods & Service terms.
- Conducted red teaming exercises to evaluate and improve the robustness of Generative AI applications against adversarial attacks.
- Implemented vector search using an Elasticsearch backend, tuned HNSW implementation to balance throughput & recall.
- Data driven refinement of our search & recommendation systems, improving relevance metrics by 20%.
- Implemented an end to end MLOps strategy to improve model deployment times to <1 day across multiple environments.
- Architected & implemented ETL pipelines using Polars, replacing an expensive Azure Databricks implementation, reducing compute costs by 50% & enabling zero downtime deployments.

Tech Stack: Azure OpenAI, Hugging Face, Azure ML Pipelines, Azure Functions, Elasticsearch, Polars

---

Incopro

Lead Machine Learning Engineer

February 2020 - November 2021

Led a team focused on transforming business operations through secure and innovative ML solutions.

- Led the design and implementation of an MLOps framework on Google Cloud Platform, using Vertex AI Pipelines to orchestrate training, evaluation, approval and deployment workflows. This improved time to market for new models to <1 day, allowing for client specific model deployments.
- Built data processing pipelines on Spark clusters running on GCP to prepare large-scale text and image datasets for model training, batch inference and recommendation workloads.
- Developed and deployed NLP and computer vision models for intellectual property protection and fraud identification, including a multilingual BERT-based model for detecting IP infringements across 90+ languages.
- Implemented a recommendation engine based on text and image similarity, enhancing the detection of IP violations.
- Mentored a software engineer to help her transition into a machine learning role.

Tech Stack: Vertex AI Pipelines, TensorFlow, PySpark, Google Cloud Platform, MLflow

Machine Learning Engineer

November 2018 - February 2020

Focused on the research and development of ML models for IP protection.

- Engineered a production system for predicting IP infringement, leveraging NLP techniques and deploying it within a secure, containerized environment. Increased number of enforced infringements by 10% per analyst (the company employed ~200 analysts) .

- Built & deployed an image similarity algorithm to identify IP infringing items. This feature was instrumental in securing a major contract with a blue-chip client, contributing to 10% of Annual Recurring Revenue (ARR).

Tech Stack: TensorFlow, FastText, Google Cloud Platform, MLFlow

---

Office for National Statistics

Machine Learning Engineer

September 2017 - November 2018

Contributed to national-scale data integration projects with a focus on secure data handling.

- Developed ML models and data pipelines to link and process government administrative data, supporting national survey efforts.
- Worked on the UK Census 2021 project, developing a highly scalable backend for the Census data collection system.

Tech stack : Python, PySpark, Cloudera

Barclays

Data Scientist

September 2018 - October 2018

Seconded to Barclays for a project focused on using payment data to create economic indicators.

- Worked on a high-impact project involving secure and ethical use of financial data to develop regional economic models.

This was a continuation of my work as part of the winning team of an ONS X Barclays Hackathon.

Purple Secure Systems

Software Engineer

August 2014 - October 2016

Engaged in full-stack development of big data processing systems for multiple clients in the government and defense sectors.

Tech stack: Java, Python, PySpark, AngularJs

Education

University of Bath

Master of Science in Applied Mathematics

2016 - 2017

Bachelor of Science in Physics

2010 - 2013

Honors & awards

Winner of ONS x Barclays Data Hackathon

Using payments data to create more granular and timely estimates for GVA, (Gross Value Added).